

Technical Note: Empirical Framework

Climate Shocks, Health Costs, and Provider Heterogeneity (Three Papers)

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This note describes the identification strategy, shock definitions, and estimation procedure for the three empirical papers in the thesis. Section 1 introduces the two data panels. Sections 2–4 cover each paper in turn. Section 5 collects inference details.

1. Two Estimation Panels

The thesis uses two distinct unit-of-observation panels that share the same county-level climate shock series but measure different outcome domains.

Panel A — County × Year (demand / consumer side) • Unit: county FIPS code × calendar year • Window: 2011–2023 (13 years) • N: ~41,400 county-years, 3,155 counties • Outcomes: HIX benchmark premium, medical-debt share, personal income per capita, civilian employment • Fixed effects: county FE + year FE • Clustering: state (nests rating areas)	Panel B — Hospital × Year (supply / provider side) • Unit: hospital CMS Certification Number (CCN) × year • Window: 2011–2023 • N: ~59,900 hospital-years, 5,119 hospitals • Outcomes: operating margin, net margin, uncompensated care (dollars, % of net patient revenue) • Fixed effects: hospital (CCN) FE + year FE • Clustering: state
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Each hospital is geo-coded to a single county (modal county per ZIP code, hospital location — not the residential catchment allocation used for consumer debt). The county-level climate shocks are then attached to the hospital by that county mapping, so the hospital panel uses the *same* shock indicators as the county panel.

1.1 Binary shock indicators

All three papers use the same four binary shock series, constructed at the county × year level:

Indicator	Definition	Source
Is_Extreme_Drought	$PDSI \leq -4$ in county c , year t (extreme drought, Palmer scale)	NOAA NCEI
High_CDD	Cooling degree-days in top quartile of county c 's historical distribution	NOAA NCEI
High_HDD	Heating degree-days in top quartile of county c 's historical distribution	NOAA NCEI

Indicator	Definition	Source
High_AQI_Max	Maximum annual AQI > 100 in county c , year t (unhealthy threshold)	EPA AQS

Shocks are denoted generically as $S_{ct} \in \{0, 1\}$, where c indexes the county and t indexes the year. For the hospital panel, a hospital i inherits the shock of its location county: $S_{it} \equiv S_{c(i),t}$.

Note on "treatment": These are *recurring binary shocks*, not one-time absorbing treatments. A county can be in drought in 2013, out in 2014–2016, back in drought in 2017, and so on. This distinguishes the design from canonical staggered-adoption DiD (e.g., Callaway–Sant’Anna): the estimator does not assume a single irreversible treatment date, and the persistence analysis below is specifically designed to exploit this entry/exit variation. As a complementary identification check, §2.5–2.6 collapse the recurring series to a *first-onset cohort* and run a clean never-exposed-control difference-in-differences around the 2012 Midwest drought.

1.2 Shocks as draws from each county's own distribution

A natural question (raised in external review) is whether these are genuine *shocks* or merely weather that is "unanticipatable given historical averages." The construction answers this directly: every shock is defined as a **departure from the individual county's own long-run distribution**, not from a national or cross-county threshold. A "shock" is therefore, by construction, an unusually high draw *relative to what that county has historically experienced*.

- **Binary indicators.** High_CDD / High_HDD mark a year in the *top quartile of county c 's own historical degree-day distribution*; Is_Extreme_Drought marks $\text{PDSI} \leq -4$ on the Palmer scale, an absolute extremity anchored to local moisture conditions.
- **Continuous z-scores.** The companion measures Z_{ct}^{Temp} and Z_{ct}^{Precip} are standardized deviations,

$$Z_{ct} = \frac{x_{ct} - \bar{x}_c^{[1990-2000]}}{s_c^{[1990-2000]}},$$

where the per-county mean $\bar{x}_c$ and standard deviation s_c are computed over a fixed **1990–2000 baseline window** that *precedes* the 2011–2023 study period. The baseline statistics are frozen: they are estimated once, before the analysis window, and never re-estimated on study-period data.

Why this addresses the "anticipation" concern. Because the yardstick is each county's own 1990–2000 distribution, a shock is an event in the tail of the distribution the county's residents, firms, and insurers would have formed expectations around. It is not a secular trend (absorbed by county fixed effects) nor a nationally common shock (absorbed by year fixed effects); it is idiosyncratic, county-specific tail realization — the very object a "draw from a distribution" framing has in mind. Anchoring to a pre-study baseline also removes any mechanical dependence between the shock definition and the outcomes measured in 2011–2023.

1.3 Handling multiple and repeated shocks

A second review question concerns counties that experience *several* shocks — different hazards in the same year, or the same hazard repeatedly. The framework already absorbs shock multiplicity through three complementary devices, so no county's multi-shock exposure is silently collapsed to a single indicator:

- **Distributed lags.** Every specification includes the contemporaneous shock plus its first and second lags ($S_{ct}, S_{c,t-1}, S_{c,t-2}$), so a shock in consecutive years contributes through multiple coefficients rather than being overwritten.
- **Compound / any-shock indicators.** A composite Any_Shock indicator (and interacted compound specifications) capture the effect of being hit by *at least one* hazard, and of co-occurring hazards, separately from the single-hazard series.
- **Cumulative dose.** The persistence analysis (§3.6) replaces the on/off indicator with a running *count of shock-years* a county has accumulated (`cumulative_dose.R`), directly modelling repeated exposure as an increasing dose rather than a sequence of independent one-off events.

2. Paper 1 — Incidence: Do Climate Shocks Raise Health Costs?

Paper 1 asks whether contemporaneous and lagged shock exposure shifts the *level* of the outcome. The specification is a standard distributed-lag two-way fixed-effects model. Sections 2.1–2.4 develop that model; §2.5–2.6 add a sharper, single-event difference-in-differences cross-check that answers the same incidence question with a never-exposed control group.

2.1 Specification

PAPER 1 ESTIMATING EQUATION (COUNTY PANEL)

$$Y_{ct} = \alpha_c + \gamma_t + \sum_{k=0}^2 \beta_k S_{c,t-k} + \mathbf{X}'_{ct} \delta + \varepsilon_{ct} \quad (1)$$

where α_c are county fixed effects, γ_t are year fixed effects, $S_{c,t-k}$ is the binary shock indicator lagged k years, and $\mathbf{X}_{ct}$ are time-varying controls (log CPI-deflated income per capita, unemployment rate, Medicaid expansion indicator). The hospital-panel specification replaces c with hospital CCN i and α_c with α_i .

How many regressions? Equation (1) is estimated **once per (outcome, shock) pair** — a single pooled regression on all ~41,400 county-year observations. The three shock terms $S_{c,t}, S_{c,t-1}, S_{c,t-2}$ all enter the right-hand side of that one regression simultaneously; there is no loop over individual years or counties.

The three terms $\beta_0, \beta_1, \beta_2$ are recovered from that single regression. They trace how the outcome responds in the shock year itself, one year after, and two years after a shock year — all identified jointly from the same estimation.

2.2 What is the comparison?

The county and year fixed effects give the estimator a precise comparison group. Suppose Maricopa County, Arizona enters extreme drought in 2012 ($S_{c,2012} = 1$, having been out of drought in 2010 and 2011).

- **County FE (α_c)** absorbs the time-invariant average level of the outcome for Maricopa County — its baseline premium cost, demographics, and healthcare market structure. This removes the comparison problem of "Arizona counties are permanently more expensive than Vermont counties."
- **Year FE (γ_{2012})** absorbs the national average change in 2012 common to all counties — ACA rollout effects, general medical inflation, any aggregate macroeconomic movement. This removes the comparison problem of "all counties' premiums rose in 2012."

After these two absorptions, $\hat{\beta}_0$ asks: *did Maricopa County's 2012 outcome deviate from its own historical average by more than the corresponding deviation for counties that were not in drought in 2012?* The control group for $\hat{\beta}_0$ is every county with $S_{c,2012} = 0$ — observed in the same year, each adjusted for its own county-level average. $\hat{\beta}_1$ uses the same logic one year displaced: the "treated" counties are those with $S_{c,2011} = 1$, and the control is counties with $S_{c,2011} = 0$, both measured in 2012 outcomes. $\hat{\beta}_2$ repeats for the 2010 drought vintage measured in 2012 outcomes.

The identifying variation is therefore counties that *switch* drought status across years. A county that is always in drought or never in drought contributes to the precision of the county FE but not to identification of $\hat{\beta}_k$; the informative observations are the within-county year-to-year changes that coincide with entering or leaving a shock state.

2.3 Cumulative effect

To see where the cumulative effect comes from, trace what happens to a single shock year in the distributed-lag structure. Suppose Maricopa County is in drought only in 2012 ($S_{c,2012} = 1$; all other years = 0). That one event appears three times in the pooled panel regression:

Panel year (t)	Which regressor carries the 2012 drought	Coefficient estimated	Meaning
2012	$S_{c,2012}$ (contemporaneous)	$\hat{\beta}_0$	Effect on the outcome in the drought year itself
2013	$S_{c,t-1} = S_{c,2012}$ (one-year lag)	$\hat{\beta}_1$	Residual effect one year after the drought
2014	$S_{c,t-2} = S_{c,2012}$ (two-year lag)	$\hat{\beta}_2$	Residual effect two years after the drought

Each row is a separate observation in the same regression. The distributed-lag model picks up the 2012 drought's contribution in each outcome year. The **cumulative effect** sums across all three:

$$\hat{\beta}_{\text{cum}} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 \quad (2)$$

This is the total change in the outcome accumulated over the drought year and the two years following — the area under the three-period impulse response. It answers: *if a county experiences one year of drought, by how much does the outcome shift in total across the contemporaneous and two subsequent years?*

Because all three $\hat{\beta}_k$ come from the same regression, the cumulative estimate is a linear combination of coefficients that share a joint covariance structure. Its variance is:

$$\widehat{\text{Var}}\left(\hat{\beta}_{\text{cum}}\right) = \sum_{j=0}^2 \sum_{k=0}^2 \widehat{\text{Cov}}\left(\hat{\beta}_j, \hat{\beta}_k\right) \quad (3)$$

computed from the state-clustered variance–covariance matrix. Ignoring the off-diagonal covariance terms (as one would if summing three separate regressions) would give the wrong standard error. The Wald z -test uses the full 3×3 block (implementation: `lincom()` in `Code/cumulative_dose.R`).

2.4 Selected headline results

Panel	Shock	Outcome	Cumulative $\hat{\beta}$	p
Hospital	High_CDD (heat)	Uncompensated care (% NPR)	+1.22 pp	0.063
Hospital	High_HDD (cold)	Uncompensated care (\$2023)	+\$1.55M per hospital	0.022
Hospital	Is_Extreme_Drought	Uncompensated care (% NPR)	−1.68 pp	0.032

Drought sign caveat: The negative drought coefficient on uncompensated care is a measurement artifact, not a causal benefit. Credit-bureau and billing measures of health-cost distress systematically undercount in drought-stricken areas (lower insurance rates, lower utilization, thinner credit files). This recurs across the demand and supply sides and is flagged throughout.

2.5 A sharper control group: the 2012 Midwest drought as a natural experiment

The distributed-lag model in §2.1 identifies $\hat{\beta}_k$ from *every* county that switches drought status — including counties that flicker in and out of drought repeatedly and counties hit in many different years. That is statistically efficient, but the control group for any given drought year ($S_{c,t} = 0$) silently includes counties that were droughted in *other* years. The committee asked for a complementary design that removes this contamination: isolate one large, well-understood event, and compare the counties it hit *for the first time* against counties that were *never* hit at all. The 2012 Midwest drought — the most geographically extensive single-year extreme drought in the 2011–2023 panel — is the natural candidate.

§2.5 ESTIMATING EQUATION — SHARP 2×2 DIFFERENCE-IN-DIFFERENCES

$$Y_{ct} = \alpha_c + \gamma_t + \tau (\text{Treated}_c \times \text{Post}_t) + \varepsilon_{ct} \quad (\text{D1})$$

$\text{Treated}_c = 1$ if county c 's *first-ever* extreme-drought year in the panel is 2012; $\text{Post}_t = 1$ for $t \geq 2012$. The county FE absorbs the time-invariant Treated_c main effect and the year FE absorbs the Post_t main effect, so only the interaction survives — $\hat{\tau}$ is the average treatment effect on the treated (ATT).

Treated vs. control. Treated = 139 counties (first-onset 2012); control = 2,534 counties that are *never* in extreme drought across all thirteen years. Counties that are droughted but whose first onset is some other year are *excluded* from this estimation — that exclusion is exactly what makes the control group clean. Estimated once per (outcome) with `feols(Y ~ Treated:Post | fips_code + Year)`, clustered at the state level.

2.5.1 What is the comparison?

The contrast with §2.2 is precise. In the distributed-lag model, $\hat{\beta}_0$'s control group was "every county not in drought *this* year" — which includes a county that was in drought last year or will be next year. Here the control group is purified to counties that are *never* droughted in the whole window. The 2012 drought cohort is then compared, before vs. after 2012, to that never-exposed pool, before vs. after 2012:

	Pre (2011)	Post (2012–2023)	Within-group change
Treated (139 first-onset-2012 counties)	$\bar{Y}_{\text{pre}}^T$	$\bar{Y}_{\text{post}}^T$	$\Delta^T = \bar{Y}_{\text{post}}^T - \bar{Y}_{\text{pre}}^T$
Control (2,534 never-exposed counties)	$\bar{Y}_{\text{pre}}^C$	$\bar{Y}_{\text{post}}^C$	$\Delta^C = \bar{Y}_{\text{post}}^C - \bar{Y}_{\text{pre}}^C$
Difference-in-differences			$\hat{\tau} = \Delta^T - \Delta^C$

The two-way FE regression (D1) is the multi-year, multi-county generalization of this 2×2 table: it replaces the single post mean with a FE-adjusted average over all post years and all 2,534 control counties, but $\hat{\tau}$ has exactly the 2×2 interpretation — *how much more did the treated counties' outcome move, relative to never-droughted counties, after the 2012 drought hit?*

2.5.2 Worked numerical example

EXAMPLE: 2012 MIDWEST DROUGHT → PERSONAL INCOME PER CAPITA (PCPI, \$2023)

The specification is equation (D1) with $Y = \text{PCPI_Real}$, estimated on the 139 treated + 2,534 never-exposed counties, $N = 34,355$ county-years, clustered at the state level (Code/run_did_analysis.R).

Step 1 — The 2×2 arithmetic

These are the *actual* sample group means (county-year averages of PCPI_Real, in \$2023). Pre = 2011 (the only pre-drought year in the panel); Post = 2012–2023. Real personal income per capita rose nationally over the decade, so *both* groups go up — the question is by how much less the treated group rose:

	Pre (2011)	Post (2012–2023 avg)	Change Δ
Treated (139 counties)	\$48,488	\$51,435	+\$2,946
Control (2,534 counties)	\$49,297	\$53,554	+\$4,257
$\hat{\tau} = \Delta^T - \Delta^C$			-\$1,311

Both groups grew, but the drought cohort grew **\$1,311 per capita less** (\$2,946 – \$4,257). That gap — not the raw post-period difference — is the causal estimate, because the control's +\$4,257 captures the national income growth the treated counties would have enjoyed absent the drought. With a single pre-period, the two-way FE regression (D1) returns this 2×2 difference exactly: $\hat{\tau} = -\$1,311$.

Step 2 — Inference

The state-clustered standard error is $\widehat{SE}(\hat{\tau}) = 576$:

$$z = \frac{\hat{\tau}}{\widehat{SE}(\hat{\tau})} = \frac{-1,311}{576} = -2.28 \quad p = 0.027$$

We reject the null of no income effect at the 5% level.

Step 3 — The employment companion

Re-running (D1) with $Y = \text{Civilian_Employed}$ on the same cohort gives a sharper result:

$$\hat{\tau} = -2,053 \text{ persons}, \quad \widehat{SE} = 478, \quad z = -4.29, \quad p = 0.0001$$

The 2012 drought cohort lost about 2,053 employed persons per county relative to never-droughted counties.

INCIDENCE CONFIRMED — income & jobs fall

The clean never-exposed design recovers a negative real-economy incidence of drought — a per-capita income decline and an employment decline — that is robust to the contamination concern in §2.2. This is the design the write-up leads with, because it has the largest control pool and the most transparent counterfactual.

Why are the medical-debt outcomes null here? In the same 2×2, $\text{Medical_Debt_Share}$ ($\hat{\tau} = -0.006$, $p = 0.49$) and $\text{Hosp_BadDebt_PerCapita}$ ($\hat{\tau} = +6.3$, $p = 0.29$) are insignificant. This is expected, not contradictory: the 2×2 averages the effect over *eleven* post-drought years, while the local-projection analysis (Paper 2) finds the debt response peaks at $h = 1-2$ and fades. A long-run average of a transient spike is close to zero. The real-economy outcomes (income, employment), by contrast, are persistent and so survive the averaging.

2.5.3 Doubly-robust check: conditional parallel trends

The sharp 2×2 assumes *unconditional* parallel trends — that, absent the drought, the treated counties' outcomes would have moved like the never-exposed pool's. That is a strong assumption here, because the two groups are compositionally different: the 139 treated counties are disproportionately rural agricultural counties of Georgia, the Mountain West, and the Plains, while the 2,534 never-exposed counties skew more urban, coastal, and eastern. If those county types were on different secular trajectories (farm economy cycles, rural population decline) for reasons unrelated to drought, the 2×2 would mistake that divergence for a treatment effect.

The frontier response is to move to *conditional* parallel trends via the Sant'Anna–Zhao (2020) **doubly-robust** DiD: it combines an outcome-regression and a propensity-score model, and is consistent if *either* is

correctly specified. We condition only on **pre-treatment baseline (2011)** covariates — log population, baseline median household income, and Census division. We deliberately do *not* condition on contemporaneous income, employment, or premiums: those are *mediators* of the climate shock, and controlling for them would be post-treatment bias (a "bad control" that absorbs part of the very effect being estimated).

§2.5.3 ESTIMATOR — SANT'ANNA–ZHAO DOUBLY-ROBUST ATT

$$\hat{\tau}_{\text{dr}} = \mathbb{E} \left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))}{\mathbb{E}[\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))]} \right) (\Delta Y - \hat{m}_0(X)) \right] \quad (\text{D2})$$

ΔY is the unit's pre-to-post outcome change (post collapsed to the 2012–2023 mean), $\hat{\pi}(X)$ the estimated propensity score, and $\hat{m}_0(X)$ the control-group outcome-change regression, all on baseline covariates X . Estimated with `DRDID::drdid` (improved/locally-efficient form) on the 2012 cohort vs. the never-exposed controls. Implementation: `Code/did_robustness/02_doubly_robust_did.R`.

Outcome	Unconditional $2 \times 2 \hat{\tau}$	Doubly-robust $\hat{\tau}_{\text{dr}}$	DR 95% CI	Verdict
PCPI per capita (\$2023)	−\$1,311	−\$1,451	[−2,461, −441]	Confirmed — stronger
Civilian employed	−2,053	−871	[−1,719, −23]	Attenuated ~58%
Median HH income (\$2023)	−\$992	−\$1,186	[−2,140, −232]	Confirmed
Medical-debt share	−0.006 (n.s.)	−0.011	[−0.019, −0.002]	Measurement-fragile

The **income** result is robust to conditioning — it is essentially unchanged, if anything slightly stronger (−\$1,451). The **employment** result is *not*: conditioning on baseline composition shrinks it by roughly 58% (from −2,053 to −871), with a confidence interval that now barely excludes zero. The income effect is therefore the credible headline; the employment effect should be read as a weaker, composition-sensitive secondary finding.

Does the 2012 effect generalize? Re-estimating the doubly-robust design as a full Callaway–Sant'Anna model across *all* drought cohorts (2011–2022, never-treated controls, same baseline covariates) yields a *null* average: PCPI +\$350 (s.e. 585), employment +2,609 (s.e. 2,245), both insignificant, and the pooled employment event-study even shows positive pre-trends. The negative real-economy effect is thus specific to the 2012 natural experiment — the largest, cleanest single event — and should not be presented as the effect of a *typical* drought cohort.

Two caveats carried throughout. (i) *Estimand*. Treatment is recurring: the 2012 cohort is actually in extreme drought only ~13% of its post-2012 county-years, so $\hat{\tau}$ is an *intent-to-treat* "effect of first drought onset," not the effect of being currently droughted. (ii) *Label*. The "2012 Midwest drought" name is historical shorthand; the extreme-PDSI cohort is concentrated in Georgia, the Mountain West, and the Plains rather than the Midwest proper.

2.6 Callaway–Sant'Anna extension: event-time profiles

The sharp 2×2 uses a single cohort and a single pre-period. A natural generalization keeps the never-exposed control group but admits *multiple* first-onset cohorts (counties whose first drought is 2011, 2012, 2013, 2021, 2022), estimates a separate ATT for each cohort against the never-treated controls, and then aggregates by event time $e = t - g$ (years since a cohort's onset year g), weighting by cohort size. This is the Callaway–Sant'Anna (2021) group-time ATT estimator.

Reconciling with "recurring shocks" (§1). The §1 note flagged that our shocks are recurring, not absorbing — which is why canonical staggered-adoption DiD is not the *primary* design. The DiD here sidesteps that by defining each cohort on its *first* onset and using only never-treated counties as controls. It deliberately trades away the entry/exit richness (which Paper 2 exploits via onset/exit states) in exchange for a clean, contamination-free counterfactual. The two designs answer different questions and are reported as complementary, not competing.

Because the local `did` R package would not build on the project toolchain, the group-time ATTs are implemented manually with `fixest::feols` (per-cohort 2×2 against never-treated, aggregated to event time) — also more consistent with the project's `fixest`-only convention. The drought cohorts and their sizes are 2011 (251), 2012 (139), 2013 (30), 2021 (50), 2022 (49). The headline aggregated cells:

Event time e	Outcome	$\widehat{ATT}_e$	p	Cohorts pooled (treated N)
$e = 0$	PCPI per capita (\$2023)	−\$1,050	0.0018	4 cohorts (268)
$e = 0$	Civilian employed	−142	0.0016	4 cohorts (268)
$e = 11$	Civilian employed	−3,915	0.0001	1 cohort — 2012 (139)

The event-time profile tells a compounding story: the contemporaneous ($e = 0$) income hit of about \$1,050 per capita matches the sharp 2×2 , and the employment effect, small on impact, deepens to roughly −3,915 jobs eleven years out — though only the large 2012 cohort survives to that horizon, so the long-run cell rests on a single cohort and is read as suggestive of persistence rather than a precise decade-out point estimate.

Parallel-trends caveat. The panel starts in 2011, so the 2012 cohort has only a single pre-period ($e = -1$) and a formal pre-trend test is not feasible for it; the diagnostic is a visual event-study plot (Analysis/plots/did/eventstudy_*.png), which shows reasonably flat pre-periods for the drought PCPI and employment outcomes. The analogous heat (CDD) DiD is *not* given headline status: CDD treatment concentrates in southern states and never-exposed controls in northern states, so two-way FE cannot rule out region-secular trends. Drought is the cleanest event and the one the write-up leads with.

How this sits beside Papers 1–2. The distributed-lag TWFE (§2.1) and the local projection (Paper 2) identify *dynamic responses to year-over-year changes* in shock status; the DiD here identifies the *persistent treated-vs-never-exposed gap*. They are complementary lenses on the same shock, and where they appear to disagree (e.g., Paper 2's "no incremental scarring" reading of cold spells vs. the DiD's long-run gap) the resolution is that each design defines a different estimand — the LP measures the response around entry and exit, the DiD measures the cost of *being* a shock-exposed county.

3. Paper 2 — Persistence: Scarring, Reversibility, and Cumulative Dose

Paper 2 asks two questions about dynamics: (A) Is the cost of a shock symmetrically reversed when the shock ends, or does it leave a lasting imprint? (B) Does repeated exposure accumulate, compressing outcomes progressively? These are answered by two complementary estimators applied to both panels.

3.1 Transition-state definitions

The first step converts the binary shock series into a *transition state* that characterizes how the county (or hospital) arrived at its current shock status. Let S_{it} and $S_{i,t-1}$ be the current and lagged shock indicator. The four possible states and their dummies are:

$(S_{it}, S_{i,t-1})$	State label	Dummy	Interpretation
(1, 0)	Onset	D_{it}^{on}	First year of a new shock spell — "just treated"
(1, 1)	Persist	D_{it}^{per}	Continuation of an ongoing spell
(0, 1)	Exit	D_{it}^{ex}	First year after a spell ends — "just de-treated"
(0, 0)	<i>Out (omitted)</i>	—	<i>Not in shock this year or last year — reference group</i>

The transition dummies are undefined when either S_{it} or $S_{i,t-1}$ is missing (these observations are dropped). All three dummies are constructed inside `make_transitions()` in `Code/run_hospital_persistence.R`.

DiD analogy: $D_{it}^{on} = 1$ is the "newly treated" group in a DiD. $D_{it}^{ex} = 1$ is the "de-treated" group — a reversal experiment. The omitted category (0, 0) is the "never-treated this period or last period" control. $D_{it}^{per} = 1$ has no clean DiD analogue; it captures the effect of being in the *interior* of a sustained treatment spell.

3.2 The local-projection estimating equation

Paper 2 uses a Jordà (2005) local-projection approach: instead of estimating a single dynamic model and propagating it forward analytically, a separate pooled panel regression is estimated for each forecast horizon h . The estimating equation at horizon h is:

PAPER 2 ESTIMATING EQUATION — ONSET/PERSIST/EXIT

$$Y_{i,t+h} = \alpha_i + \gamma_t + \beta_h^{on} D_{it}^{on} + \beta_h^{per} D_{it}^{per} + \beta_h^{ex} D_{it}^{ex} + \varepsilon_{i,t+h} \quad (4)$$

The left-hand side is the *level* of Y measured h years after the transition year t . The transition dummies on the right-hand side are always indexed by t — the year in which the county changed shock state. Unit and year fixed effects (α_i, γ_t) absorb time-invariant heterogeneity and common time trends, with γ_t indexed by the transition year t , not the outcome year $t + h$.

How many regressions? Equation (4) is estimated **three times per (outcome, shock) pair** — once for $h = 0$, once for $h = 1$, once for $h = 2$ — each time as a single pooled panel regression over all county-year (or hospital-year) observations. There is no loop over years or counties; all units and all years enter each regression together.

The three regressions differ only in their left-hand side and their usable sample:

Horizon h	Left-hand side	Right-hand side transition dummies	Sample lost
$h = 0$	$Y_{i,t}$ (outcome in the transition year)	$D_{it}^{on}, D_{it}^{per}, D_{it}^{ex}$	None (~41,400 obs.)
$h = 1$	$Y_{i,t+1}$ (outcome one year later)	$D_{it}^{on}, D_{it}^{per}, D_{it}^{ex}$	Last year of panel (~38,200 obs.)
$h = 2$	$Y_{i,t+2}$ (outcome two years later)	$D_{it}^{on}, D_{it}^{per}, D_{it}^{ex}$	Last two years of panel (~35,000 obs.)

Each of the three regressions yields its own set of coefficient estimates ($\hat{\beta}_h^{on}, \hat{\beta}_h^{per}, \hat{\beta}_h^{ex}$) and its own clustered variance–covariance matrix. The symmetry test in §3.4 is performed separately at each horizon using that horizon's vcov.

3.3 Interpretation of the three coefficients

All coefficients in equation (4) are measured relative to the reference group (0, 0):

- $\hat{\beta}_h^{on}$: the average change in $Y_{i,t+h}$ for a unit in its *first* shock year, relative to a unit that had no shock in either t or $t - 1$. This is the "treatment onset" effect h periods forward.
- $\hat{\beta}_h^{per}$: the effect for units still *inside* an ongoing spell. Relative to the onset coefficient, it tells you whether staying in the shock is more or less damaging than just entering it.
- $\hat{\beta}_h^{ex}$: the effect in the *first year after* the shock ends. If the shock were fully transitory and reversible, $\hat{\beta}_h^{ex}$ would be zero — the "de-treated" units would look identical to the "never-treated" control. A nonzero $\hat{\beta}_h^{ex}$ means the exit period differs from the reference even after the shock has cleared.

3.3a What is the comparison for each transition state?

The unit and year FEs work the same way as in Paper 1: they absorb each county's (or hospital's) permanent level and the common year trend, leaving within-unit deviations as the variation. The key addition is that the

comparison group is now *defined by transition state*, not just by the contemporaneous shock value.

Onset ($D_{ct}^{on} = 1$, i.e., $S_{ct} = 1$, $S_{c,t-1} = 0$). Suppose Maricopa County, AZ is not in drought in 2011 but enters drought in 2012. Its 2012 transition state is Onset. For horizon $h = 0$, $\hat{\beta}_0^{on}$ compares Maricopa's 2012 outcome — as a within-county deviation from its own average — to the within-county deviations of all counties whose transition state in 2012 is the reference (0, 0): not in drought this year or last year. Counties in the Persist or Exit state in 2012 are in the regression but assigned to different dummies; they do not contaminate the Onset estimate.

Persist ($D_{ct}^{per} = 1$, i.e., $S_{ct} = 1$, $S_{c,t-1} = 1$). Suppose Maricopa remains in drought in 2013 as well ($S_{c,2013} = 1$). Its 2013 transition state is Persist. $\hat{\beta}_0^{per}$ compares Maricopa's 2013 within-county deviation to the same (0, 0) reference counties in 2013. This answers: how does the *interior* of a sustained drought spell differ from never being in drought — in the same year, after removing each unit's permanent level?

Exit ($D_{ct}^{ex} = 1$, i.e., $S_{ct} = 0$, $S_{c,t-1} = 1$). Suppose drought clears in 2014 ($S_{c,2014} = 0$, $S_{c,2013} = 1$). Maricopa's 2014 transition state is Exit. $\hat{\beta}_0^{ex}$ compares Maricopa's 2014 within-county deviation to (0, 0) reference counties in 2014. The drought is *gone* in 2014 — if $\hat{\beta}_0^{ex} \neq 0$, it means the county does not immediately revert to its pre-shock trajectory even though the shock has cleared. That is what scarring means empirically.

One regression, three simultaneous comparisons. All three dummies enter a *single* regression at each horizon h , so the reference (0, 0) group is the common benchmark for Onset, Persist, and Exit simultaneously. The regression partitions the county-year observations by which of the four transition states they occupy in year t ; the (0, 0) category is the omitted group. Year FEs absorb any common shock in t that would otherwise confound the comparison.

3.4 The symmetry (scarring) test

"Reversibility" has a precise formulation. If a shock is *fully reversible*, the cost of entering (relative to never being hit) should be exactly undone on exit. The onset and exit coefficients would then be equal in magnitude but opposite in sign, so their sum is zero. The formal null is:

$$H_0 : \beta_h^{on} + \beta_h^{ex} = 0 \quad (\text{fully reversible}) \quad (5)$$

Define the **asymmetry statistic**:

$$\hat{L}_h = \hat{\beta}_h^{on} + \hat{\beta}_h^{ex} \quad (6)$$

Because both coefficients come from the *same* regression at horizon h , their covariance is estimable from the single fitted model's variance–covariance matrix. The standard error of $\hat{L}_h$ is:

$$\widehat{SE}(\hat{L}_h) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_h^{on}) + \widehat{\text{Var}}(\hat{\beta}_h^{ex}) + 2\widehat{\text{Cov}}(\hat{\beta}_h^{on}, \hat{\beta}_h^{ex})} \quad (7)$$

The test statistic is:

$$z_h = \frac{\hat{L}_h}{\widehat{\text{SE}}(\hat{L}_h)} \xrightarrow{d} \mathcal{N}(0, 1) \quad (8)$$

under H_0 , using state-clustered standard errors throughout.

Interpretation of $\hat{L}_h$:

- $\hat{L}_h = 0$: onset and exit are mirror images — the shock is fully reversible.
- $\hat{L}_h \neq 0$ and we reject H_0 : the dynamics are **asymmetric**.
- The *sign* of $\hat{L}_h$ and the signs of the individual coefficients together tell the economic story (scarring vs. over-recovery vs. a non-monotone path).

Why not run two separate regressions? Estimating onset and exit in separate models would require an assumption about independence (or use a bootstrap). The joint specification recovers $\widehat{\text{Cov}}(\hat{\beta}^{on}, \hat{\beta}^{ex})$ directly from the model, so the test is a standard Wald linear combination — no additional assumption beyond the regression itself.

3.5 Worked numerical example

EXAMPLE: EXTREME DROUGHT → HOSPITAL OPERATING MARGIN, HORIZON $H=0$

The specification is equation (4) with $Y = \text{Hosp_OperatingMargin}$, $\text{shock} = \text{Is_Extreme_Drought}$, horizon $h = 0$. The model is:

$$\text{OpMargin}_{i,t} = \alpha_i + \gamma_t + \beta_0^{on} D_{it}^{on} + \beta_0^{per} D_{it}^{per} + \beta_0^{ex} D_{it}^{ex} + \varepsilon_{it}$$

estimated by `feols` (Correia 2016, *fixest*), $N = 57,302$ hospital-years, clustered at the state level.

Step 1 — Point estimates

Coefficient	Estimate	Meaning
$\hat{\beta}_0^{on}$	+0.01586	First drought year vs. (0,0): margin 1.59 pp higher
$\hat{\beta}_0^{per}$	-0.00168	Continued drought year vs. (0,0)
$\hat{\beta}_0^{ex}$	+0.01089	First post-drought year vs. (0,0): margin still 1.09 pp higher

Step 2 — Asymmetry statistic

If the shock were fully reversible, the onset coefficient (+1.59 pp) should be undone on exit:

$\hat{\beta}_0^{ex} \approx -0.0159$ and $\hat{\beta}_0^{on} + \hat{\beta}_0^{ex} \approx 0$. Instead:

$$\hat{L}_0 = \hat{\beta}_0^{on} + \hat{\beta}_0^{ex} = 0.01586 + 0.01089 = 0.02675$$

Both coefficients are positive: entering a drought *and* exiting a drought are each associated with higher margins relative to the never-shocked reference. The exit state does not "unwind" the onset — it

remains elevated.

Step 3 — Standard error from the joint vcov

Let V be the 3×3 state-clustered variance–covariance matrix for $(\hat{\beta}_0^{on}, \hat{\beta}_0^{per}, \hat{\beta}_0^{ex})$. We need only the $(1, 1)$, $(3, 3)$, and $(1, 3)$ entries:

$$\widehat{SE}(\hat{L}_0) = \sqrt{V_{11} + V_{33} + 2V_{13}} = 0.00981$$

(reported directly by `transition_symmetry_test()` in `Code/transition_symmetry.R`).

Step 4 — Test

$$z_0 = \frac{0.02675}{0.00981} = 2.73 \quad p = 2 \times \Phi(-2.73) \approx 0.0064$$

We reject $H_0 : \beta_0^{on} + \beta_0^{ex} = 0$ at the 1% level.

ASYMMETRIC — scarring

The drought shock leaves a net imprint: even after drought conditions clear, the hospital's operating margin does not revert to the pre-shock reference level. This pattern (positive onset *and* positive exit, both relative to never-shocked) is consistent with survivorship and financial adaptation — hospitals that pass through a drought spell and remain in the panel are systematically different from hospitals that never face such shocks — rather than straightforward financial deterioration. Uncompensated care follows a parallel asymmetric pattern with negative coefficients throughout (both entry and exit associated with *lower* measured uncompensated care, a measurement-fragility finding discussed under Paper 1).

3.6 Cumulative dose (second persistence lens)

The onset/exit specification in §3.2–3.5 isolates the dynamics *around a single spell* — what happens at entry and immediately after exit. A separate question is whether the *total stock* of climate exposure accumulated over a unit's history has a monotone dose-response relationship with outcomes: does a hospital (or county) with ten shock-years behind it look systematically different from one with two? This is the "wear and tear vs. adaptation" test.

Building the dose counter

For each unit i , define the running count of shock-positive years up to and including year t :

$$C_{it} = \sum_{s \leq t} S_{is} \tag{9}$$

The counter increments by 1 in every shock year and **never resets** when the unit exits and re-enters the shock state — it measures total accumulated exposure, not current exposure.

Concretely, suppose Hospital H is in a county that experiences the following `Any_Climate_Shock` sequence (drought *or* extreme heat *or* extreme cold in the hospital's county):

Year	S_{it} (any shock this year)	C_{it} (cumulative shock-years to date)	Dose bin
2011	0	0	0 (reference)
2012	1	1	1–3
2013	1	2	1–3
2014	0	2	1–3 ← counter does not reset on exit
2015	0	2	1–3
2016	1	3	1–3
2017	1	4	4–6
2018	0	4	4–6
2019	1	5	4–6
2020	1	6	4–6
2021	0	6	4–6
2022	1	7	7–9
2023	1	8	7–9

Each row is one observation in the panel regression. The regressor C_{it} varies within hospital i across years — it is this within-hospital variation in accumulated dose that identifies the dose-response, after the hospital FE absorbs Hospital H's permanent (time-invariant) level of, say, operating margin.

Two estimating equations

Specification A — Linear + quadratic dose-response (one regression per outcome):

§3.6 SPEC A — QUADRATIC DOSE-RESPONSE

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon_{it} \quad (10)$$

Estimated once per outcome on the full panel. β_1 and β_2 are recovered jointly. The marginal effect of one additional accumulated shock-year, evaluated at dose level x , is:

$$\left. \frac{\partial Y}{\partial C} \right|_{C=x} = \beta_1 + 2\beta_2 x$$

This is a linear combination of $\hat{\beta}_1$ and $\hat{\beta}_2$ with weights $(1, 2x)$, tested via the `lincom()` Wald procedure using the joint vcov. For Hospital H above, evaluating at $x = 5$ cumulative shock-years gives the marginal effect on operating margin of gaining one more shock-year when already at five years of prior exposure.

Reported result. For operating margin, the estimated marginal effect at $x = 5$ is $+0.0023$ ($p = 0.010$); at $x = 10$ it shrinks to $+0.0018$ ($p = 0.14$). The positive sign at moderate dose indicates that hospitals which have accumulated more shock-years are not deteriorating — the likely reading is survivorship (financially weaker hospitals exit the panel first) or operational adaptation.

Specification B — Dose bins (one regression per outcome):

§3.6 SPEC B — BINNED DOSE

$$Y_{it} = \alpha_i + \gamma_t + \delta_{1-3} \mathbf{1}[C_{it} \in 1-3] + \delta_{4-6} \mathbf{1}[C_{it} \in 4-6] + \delta_{7-9} \mathbf{1}[C_{it} \in 7-9] + \delta_{10+} \mathbf{1}[C_{it} \geq 10] + \varepsilon_{it} \quad (11)$$

Reference category: $C_{it} = 0$ (no prior shock exposure). Each $\hat{\delta}$ is the average within-unit deviation in Y for observations in that dose bin, relative to zero-exposure observations in the same year (after year FE removal).

Hospital H contributes to the 1–3 bin in 2012–2016, to the 4–6 bin in 2017–2021, and to the 7–9 bin in 2022–2023.

The key contrast is between the highest and lowest non-zero bins — a high-vs-low test of whether long-exposed units differ from barely-exposed units:

$$\hat{\Delta} = \hat{\delta}_{10+} - \hat{\delta}_{1-3} \quad (12)$$

computed via `lincom` from the same regression's joint `vcov`. **Reported result:** for operating margin, $\hat{\Delta} = +0.018$ ($p = 0.033$) — hospitals in the 10+ bin outperform hospitals in the 1–3 bin by 1.8 pp on operating margin, again consistent with survivorship rather than accumulating harm.

Survivorship caveat. The panel is unbalanced: hospitals that accumulate many shock-years and remain in the 2011–2023 window are a selected sample of financially viable survivors. The positive dose-response on operating margin does *not* mean climate exposure improves hospital finances; it means the hospitals still in the data after repeated shocks are the resilient ones. This is why the persistence story is anchored to the **drought hysteresis result** from §3.4–3.5 (which does not require a long accumulation window), not to the dose-response as a primary claim.

4. Paper 3 — Inequality and Provider Heterogeneity

Paper 3 asks whether the climate-shock effects documented in Papers 1 and 2 are distributed evenly across the population and provider base, or concentrated in already-disadvantaged groups. The demand side uses county-level social vulnerability (CDC SVI) as the moderator; the supply side uses hospital-level characteristics.

4.1 Specification

PAPER 3 ESTIMATING EQUATION — INTERACTION / HETEROGENEITY

$$Y_{it} = \alpha_i + \gamma_t + \beta S_{it} + \theta(S_{it} \times M_{it}) + \phi M_{it} + \mathbf{X}_{it}'\delta + \varepsilon_{it} \quad (10)$$

where M_{it} is a moderator variable (binary or continuous). β is the shock effect for units with $M = 0$. θ is the *differential* shock effect for high- M relative to low- M units. The total marginal shock effect at moderator level m is $\beta + \theta \cdot m$.

4.2 Moderators

County panel — demand side

- **SVI quartile** (CDC/ATSDR Social Vulnerability Index, overall and four themes: socioeconomic status, household characteristics, racial/ethnic minority status, housing/transportation type)
- Quartile 4 = most vulnerable counties; interaction dummy = $\mathbf{1}[SVI_{ct} \geq Q_4]$

Hospital panel — supply side

- **SafetyNet**: top-quartile Medicaid + uncompensated payer mix
- **Ownership**: Non-Profit / For-Profit / Government (two dummies)
- **MedicaidExpansion**: state-year ACA expansion status
- **MarketConcentration**: county-year HHI from NPR shares by health system (HighConcentration ≥ 0.25)

4.3 Reading the marginal effects

The coefficient of interest is $\hat{\theta}$: does the shock effect differ between high- M and low- M units? The two marginal effects are:

Group	Marginal shock effect	Standard error
Low M ($M = 0$)	$\hat{\beta}$	$\widehat{SE}(\hat{\beta})$
High M ($M = 1$)	$\hat{\beta} + \hat{\theta}$	$\sqrt{V_{\beta\beta} + V_{\theta\theta} + 2V_{\beta\theta}}$
Differential	$\hat{\theta}$	$\widehat{SE}(\hat{\theta})$

The high- M marginal effect is again a linear combination from the same model's vcov. The verdict distinguishes two directions:

- For outcomes where a *larger value* = more strain (e.g., uncompensated care %), a positive $\hat{\theta}$ ($\hat{\theta} > 0$) means **strain concentrates in high- M units**.
- For outcomes where a *more-negative value* = more strain (e.g., operating margin), a negative $\hat{\theta}$ means concentration in vulnerable units.

4.4 Selected results

Shock × Moderator	Outcome	$\hat{\theta}$ direction	p	Verdict
High_CDD × SafetyNet	Uncomp. care % NPR	Higher for SafetyNet	0.021	Concentration in safety-net
Drought × SVI (county)	Personal income p.c.	Larger loss in high-SVI	<0.05	Amplified in vulnerable counties
High_HDD × SafetyNet	Uncomp. care % NPR	Higher for non-safety-net	0.011	Reversed (non-SN)
Drought × MedicaidExpansion	Uncomp. care % NPR	Higher in expansion states	<0.001	Expansion state (surprising)

The key headline is that *heat*-driven uncompensated care concentrates in safety-net hospitals (the supply-side analogue of high-SVI county amplification on the demand side). However, interactions are shock- and outcome-specific: not all shock-moderator combinations point in the "vulnerability amplifies" direction, and results are reported transparently.

5. Inference Details

5.1 Variance estimator

All standard errors are clustered at the **state level** throughout (county panel: ~ 50 clusters; hospital panel: ~ 50 clusters). State-level clustering is conservative relative to county-level clustering and accounts for within-state correlation in both the shock series and the outcome series. For HIX premium outcomes specifically, the county panel also reports rating-area-clustered SEs as a robustness check, since counties sharing a rating area have identical premiums by construction.

5.2 Linear combinations and Wald tests

All post-estimation quantities — cumulative distributed-lag effects (equation 2–3), asymmetry statistics (equation 6–7), marginal effects at moderator values (Section 4.3), and dose-level marginal effects — are computed via the `lincom()` helper (weights vector $\mathbf{w}$):

$$\hat{L} = \mathbf{w}'\hat{\beta}, \quad \widehat{\text{Var}}(\hat{L}) = \mathbf{w}'\hat{V}\mathbf{w} \quad (11)$$

where $\hat{V}$ is the state-clustered variance–covariance matrix of the full coefficient vector $\hat{\beta}$. The z -statistic uses the normal approximation; all hypothesis tests are two-tailed at the $\alpha = 0.05$ level unless noted.

5.3 Estimation software

All models are estimated using `fixest::feols` (Correia, Zylkin, and Bergé, 2020), which implements the within-group demeaning for two-way fixed effects and exports the full clustered `vcov` for the post-estimation

lincom steps. Interaction models use `feols(Y ~ S * M | unit + Year)`, which correctly includes main effects, the interaction term, and absorbs both fixed-effect dimensions simultaneously.

5.4 Panel structure and observation counts

Panel	Unit FE	Year FE	Clusters	Approx. <i>N</i>
County × year	3,155 county FE	13 year FE	~50 states	41,400
Hospital × year	5,119 hospital (CCN) FE	13 year FE	~50 states	59,900

Horizon regressions ($h = 1, 2$) lose the last h years of the panel on the left-hand side, reducing N by roughly $h/13$ per horizon. The shock lag in Paper 1 ($k = 1, 2$) analogously drops the first k years on the right-hand side.

6. Mechanisms: What Explains the Reduced-Form Results?

The papers above establish credible reduced-form relationships between climate/pollution shocks and health-cost outcomes. External review asked the natural follow-up: *through what channels*, and in particular, how much of the effect is the obvious **agricultural income channel** (drought/cold devastate farming → depress local income → drive the premium/debt/employment results) versus something else? This section summarizes the mechanism analysis (full detail in `Analysis/mechanism/mechanism_channels.md` — a literature-grounded channel map — and `Analysis/mechanism/mechanism_verdict.md` — the in-panel estimates).

6.1 The separability test

We bound the agricultural channel with a structural, pre-treatment moderator and a subsample reading. Agricultural dependence is measured as a *time-invariant* county attribute (USDA ERS 2015 farming-dependent typology; baseline 2001–2010 farm-earnings share) — never contemporaneous farm income, which would be a bad control on the causal path. For each (outcome, shock) pair we estimate an interaction $S_{ct} \times Ag_c$ (does the effect *load* on farm counties?) and re-estimate within the **bottom tercile of agricultural dependence** (urban/service counties, where a surviving effect is by construction *not* agricultural). A parallel moderator — the share of employment in climate-exposed *non-farm* industries (construction, mining, manufacturing, transport, utilities), from ACS C24030 — tests whether the residual is **broad labor exposure** rather than cropland (Graff Zivin–Neidell; Somanathan et al.; the Deryugina–Hsiang weekday-only US income signature).

6.2 Verdict — agriculture is one channel, not the channel

- **A morbidity / healthcare-utilization channel, entirely non-agricultural, is the cleanest-identified result.** Regressing county Medicare outcomes (CMS Geographic Variation PUF, 2014–2023)

on the shocks: heat raises standardized spending (+\$112 to +\$177 per beneficiary, $p < 0.02$) and ED visits (+7.8 to +9.5 per 1,000); cold raises both at a two-year lag; and **poor air quality raises ED visits** (+4.8 / +3.3 / +2.8 per 1,000, all significant) — an in-panel reproduction of Deryugina et al. (2019). None of this passes through farm income.

- **It is labor exposure, not cropland.** The cold→employment effect *survives and strengthens* in the least-agricultural counties (bottom-ag-tercile −2,011 jobs, $p = 0.05$), and the heat→employment effect loads on the climate-exposed non-farm employment share ($S \times \text{Labor}$: −689, $p = 0.009$).
- **Energy burden is a distinct distributional channel.** Heat damage concentrates in high-energy-burden counties (DOE LEAD; $S \times \text{burden}$ −1,380 jobs, $p < 0.001$). Notably it is only weakly correlated with the social-vulnerability index ($r = 0.11$), so it adds an independent margin rather than restating the high-SVI result.
- **The agricultural channel is real but narrower.** Clean farm-dependence interactions are noisy in the recurring-treatment panel; consistent with the DiD-robustness track (§2.5), the drought→income effect is *event-specific* (2012 cohort) rather than a general "droughted-county" law. Part of the drought scar is also **population selection**: drought raises net out-migration the following year (IRS SOI flows, −0.0021, $p = 0.05$).

One-paragraph answer to the reviewer. Most of what the reduced form picks up is *not* the agricultural income channel. The health-cost/utilization response is measured directly in Medicare data and is agriculture-free; the employment response is broad labor exposure that survives in non-farm counties; and the distributional amplification runs through energy burden as much as through farming. Agriculture contributes a real but event-specific, partly selection-driven drought effect — a bound on, not the whole of, the phenomenon.

Appendix: Quick-Reference Summary

Paper	Estimating equation	LHS	Key quantity
1 — Incidence	$Y_{it} = \alpha_i + \gamma_t + \sum_{k=0}^2 \beta_k S_{i,t-k} + \varepsilon$	Level of outcome in year t	Cumulative $\hat{\beta}_{cum} = \sum_k \hat{\beta}_k$ (Wald)
1-DiD — Incidence (natural experiment)	$Y_{ct} = \alpha_c + \gamma_t + \tau (\text{Treated}_c \times \text{Post}_t) + \varepsilon$	Level of outcome in year t	ATT $\hat{\tau} = \Delta^T - \Delta^C$; CS event-time $\widehat{\text{ATT}}_e$
2A — Persistence (onset/exit)	$Y_{i,t+h} = \alpha_i + \gamma_t + \beta^{on} D^{on} + \beta^{per} D^{per} + \beta^{ex} D^{ex} + \varepsilon$	Level of outcome h years forward	Asymmetry $\hat{L}_h = \hat{\beta}^{on} + \hat{\beta}^{ex}$ (Wald, joint vcov)

Paper	Estimating equation	LHS	Key quantity
2B — Persistence (dose)	$Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon$	Level of outcome in year t	Marginal effect at dose x : $\hat{\beta}_1 + 2\hat{\beta}_2 x$ (Wald)
3 — Heterogeneity	$Y_{it} = \alpha_i + \gamma_t + \beta S_{it} + \theta(S_{it} \times M_{it}) + \phi M_{it} + \varepsilon$	Level of outcome in year t	Differential effect $\hat{\theta}$; marginal at $M = 1$: $\hat{\beta} + \hat{\theta}$

Prepared for committee review · June 2026 · All estimation in R (fixest package). Shock construction:
Code/process_county_climate.R, Code/process_hospital_panel.R. Persistence helpers: Code/transition_symmetry.R,
Code/cumulative_dose.R. Natural-experiment DiD: Code/run_did_analysis.R.